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Do AI chatbots provide reliable information on Pediatric first aid? A comparative evaluation of large language models

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Abstract

Purpose: Childhood accidents are among the leading causes of injury during early childhood. This study aimed to evaluate and compare the accuracy, clarity, and comprehensiveness of pediatric first aid information generated by LLMs.

Methods: A cross-sectional comparative evaluation design was employed. Twenty standardized pediatric first aid questions were developed based on international guidelines and expert consensus. Responses generated by ChatGPT, Claude, Gemini, and Copilot were independently evaluated by a pediatric nurse and a physician using a 5-point Likert scale. Inter-rater reliability was assessed using Cohen's kappa coefficient, and differences among models were analyzed using one-way analysis of variance.

Results: Moderate inter-rater agreement was observed across all evaluation domains. Statistically significant differences were identified among the four LLMs. Claude demonstrated the highest overall performance across all evaluation domains. Gemini demonstrated relatively high accuracy but lower clarity and comprehensiveness scores. Copilot performed well in clarity but showed limited depth of clinical content. ChatGPT received the lowest scores across all assessed domains.

Conclusions: The findings reveal considerable variability in the quality of pediatric first aid information generated by LLMs. While certain models may serve as supportive educational tools, none should be considered a substitute for professional medical assessment or emergency care.

1. Introduction

1.1. Childhood injuries and public health burden

Childhood, particularly the preschool years, is a period of high risk for accidents and traumatic events. Childhood injuries account for more than 40% of all injuries, with the home environment being the most frequently reported setting where these incidents occur. This vulnerability is largely attributable to children's developmental characteristics, including limited risk perception, high mobility, and insufficient motor coordination.

Research confirms that home-related injuries are common among young children, with studies reporting injury rates ranging between 46.3% and 60% among parents of children aged 0–6 years [1,2]. The most common types of injuries include falls, bumps and abrasions, cuts, and burns, often occurring during free play in living rooms and kitchens [1,3].

1.2. Parental knowledge and health literacy in first aid

Despite their frequency, parental knowledge and self-efficacy in first aid remain insufficient. Wei et al. (2013) found that although 72% of parents demonstrated basic first aid knowledge, only 26.6% had full confidence in their abilities, with particularly low scores in drowning and CPR management [4]. Similarly, Al-Bshri and Jahan (2021) reported that 58.2% of mothers had only moderate knowledge of first aid, with inappropriate practices especially noted in burn and drowning cases [1]. Inadequate first aid responses at home not only delay appropriate care but may also contribute to preventable morbidity in children. These findings highlight the critical need for improved first aid training programs and awareness campaigns targeting parents and early childhood education professionals [3].

The literature further indicates that parents often fail to apply correct measures in emergencies, and inappropriate practices may escalate minor injuries into serious health problems [5]. From a public health perspective, such errors place an additional burden on emergency healthcare services. This risk is exacerbated in populations with low

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health literacy. For instance, a study in Canada found that 43.7% of caregivers presenting to pediatric emergency departments had low health literacy, and these individuals were significantly more likely to use emergency services for non-urgent conditions [6]. Low health literacy has also been associated with difficulties in understanding written and digital health information, particularly in time-sensitive situations such as first aid.

1.3. Artificial intelligence and large language models in healthcare

In recent years, artificial intelligence (AI) and particularly large language models (LLMs) have emerged as innovative tools in healthcare. Models such as ChatGPT, Claude, Gemini, and Copilot can respond to user queries in natural language and are rapidly becoming prevalent in both academic and everyday contexts. These systems are increasingly used by parents as informal sources of health information due to their accessibility, speed, and perceived authority.

Recent studies evaluating the performance of LLMs in healthcare have revealed both promising potential and significant limitations. For example, in pediatric chronic kidney disease information, Gemini achieved the highest quality scores compared to ChatGPT and Copilot, although all models showed limitations relative to reference sources [7]. In pediatric orthopedic conditions, ChatGPT and Gemini demonstrated moderate alignment with clinical guidelines, while concerns were raised regarding the accuracy of referenced content [8]. A systematic review further emphasized recurring ethical concerns such as bias, misinformation, and lack of transparency, reinforcing the need for human oversight [9]. These concerns are particularly critical in pediatric contexts, where incorrect guidance may directly affect child safety.

1.4. Evidence gaps in LLM evaluation in pediatric care

Although studies have begun evaluating LLMs in healthcare contexts, significant gaps remain in pediatric applications and in the development of comprehensive evaluation frameworks [10–13]. Mensah et al. (2024) highlighted the importance of context-specific validation in medical emergency scenarios [11], while Mondillo et al. (2025) reported that LLM performance tends to be weaker in pediatric contexts compared to adult medicine [12]. Similarly, a systematic review by Shool et al. (2025) indicated that most studies focus on general-purpose evaluation metrics, with limited attention to domain-specific clinical performance and caregiving contexts [13]. Notably, few studies have examined LLM performance through a nursing or caregiver-centered lens.

To date, systematic comparisons of LLM performance in simple pediatric first aid scenarios remain limited. Moreover, existing research has largely focused on single-dimensional outcomes such as accuracy, while multidimensional aspects including clarity and comprehensiveness have been underexplored. Given that parents require not only accurate but also actionable and understandable guidance, these dimensions are essential for real-world applicability.

1.5. Study aim and research questions

The aim of this study is to systematically evaluate the responses of different large language models (ChatGPT, Claude, Gemini, Copilot) to questions concerning common first aid situations in childhood. By adopting a multidimensional expert-based evaluation approach, this study seeks to address existing gaps in the literature. The findings are expected to clarify the role of AI in parental guidance and pediatric nursing practice and to provide evidence for the safe integration of such tools into healthcare-related decision support.

Research questions

- Do large language models (ChatGPT, Claude, Gemini, Copilot) differ in terms of accuracy, clarity, and comprehensiveness when responding to common pediatric first aid questions?
- According to expert evaluations, which large language model demonstrates the most reliable and consistent performance in child health and parental guidance contexts?

2. Methods

2.1. Study design

This study was designed as a structured and comparative evaluation of the responses provided by large language models (LLMs) to common first aid questions related to childhood. Data consisted of responses generated by different AI-based chatbots between 2024 and 2025.

2.2. Data source

The study was based on the analysis of responses produced by various LLMs to common first aid scenarios encountered during childhood. ChatGPT-4 (OpenAI), Claude (Claude.ai), Gemini (Google), and Copilot (Bing, Microsoft) were selected due to their accessibility, advanced language generation capabilities, and widespread use in both academic and social contexts. A set of 20 standardized first aid questions (e.g., minor cuts, burns, nosebleeds, choking) were developed, and each model was prompted independently to provide responses. Data collection was conducted between 2024 and 2025. Inclusion of multiple models allowed for a more comprehensive evaluation of AI outputs in terms of accuracy, completeness, and safety, as well as the identification of potential inter-model differences. All responses were stored securely in encrypted files and coded to anonymize AI model identity for blinded evaluation.

2.3. Sample scope and question development

The study focused on the most common first aid situations in childhood. For this purpose, 20 standardized questions were developed, guided by the *American Red Cross First Aid Guidelines*, *American Academy of Pediatrics (AAP) First Aid Recommendations*, and *World Health Organization (WHO) Child Injury and First Aid Guidelines*, as well as expert input from pediatric nurses and child health specialists. Additionally, frequently discussed first aid topics in parenting communities on platforms such as Reddit (e.g., r/Parenting, r/toddlers, r/BabyBumps) and Facebook parent groups were considered to enrich the content.

Questions were designed to be clear, simple, and representative of emergencies commonly faced by parents (e.g., minor cuts, burns, nosebleeds, choking, allergic reactions, head injuries). Each question was reviewed and revised by a pediatric nurse specialist and a physician to ensure accuracy and clarity. All questions were asked only once, from the same IP address and stable fiber internet connection. When multiple response versions were available, the first response was selected, and when conversation style modes were offered, the standard/default mode was used. For each model, a new user account was created, and each question was asked in a separate chat window. The set of 20 standardized questions was pilot-tested with 2 independent pediatric professionals to ensure clarity and relevance before full data collection (Supplementary file Section A).

Inclusion Criteria

✓ Responses generated between 2024 and 2025.
✓ Only responses from the selected AI chatbots (ChatGPT, Claude, Gemini, Copilot).
✓ Responses evaluated for scientific accuracy, safety, completeness, and practical applicability by a pediatric nurse and a physician.
✓ Responses to the predetermined 20 questions included in the analysis.
✓ Models that were web-based, accessible to the general public, up-to-date, supported multiple languages, and capable of text-based natural language processing.

Exclusion Criteria

✓ Incomplete, superficial, or vague responses.
✓ Responses containing misleading, unsafe, or clinically irrelevant recommendations.
✓ Responses inconsistent with current pediatric first aid guidelines.
✓ Inconsistent or duplicated answers when the same question was re-asked.
✓ Models that were outdated, discontinued, or with obsolete knowledge cut-off dates.
✓ Narrow or single-function dialogue systems.
✓ Systems not accessible to the public or still in the experimental phase.
✓ Models with non-transparent data security or ethical policies.

2.4. Large language model specifications

The large language models (LLMs) evaluated in this study included ChatGPT (GPT-4–based model, OpenAI), Claude (Anthropic), Gemini (Google), and Copilot (Microsoft Bing). All models were accessed through their publicly available web-based interfaces between 2024 and 2025. The default system settings provided by each platform were used, and no manual adjustment was made to parameters such as temperature, creativity mode, or response length.

Model Knowledge Cutoff:

- ChatGPT–4 (knowledge cutoff: 2024–03),
- Claude (knowledge cutoff: 2024–05).
- Gemini (knowledge cutoff: 2024–06).
- Copilot (knowledge cutoff: 2024–02).

To minimize contextual and user-history bias, a newly created user account was used for each model, and no prior interaction history was present. All questions were submitted in English using identical phrasing and structure. No follow-up prompts, clarification requests, or iterative refinements were allowed. Only the first complete response generated by each model was recorded and included in the analysis. All models were prompted using the same standardized question set. Each question was entered as a single, standalone prompt without any additional instructions, contextual framing, role assignment, or follow-up interaction. No system-level prompts or safety overrides were applied. The prompts were limited to plain text questions reflecting common pediatric first aid scenarios encountered by parents. The complete list of the 20 standardized prompts used in this study is provided below in the exact wording submitted to each model.

2.5. Data collection process

In this study, the most common first aid situations in children were identified, and corresponding questions were presented to four different AI-based language models: ChatGPT, Claude, Gemini, and Copilot. The responses from each model were recorded in written text format and constituted the primary data source of the research.

To ensure methodological rigor and reduce stochastic variability inherent in large language model outputs, each question was asked on three separate days, and consistency across responses was evaluated. Only responses demonstrating high semantic similarity and consistent clinical content were included in the analysis (Supplementary file Section C). In cases of minor variation, the most comprehensive and guideline-consistent response was selected.

To reduce subjectivity in response selection, repeated outputs were evaluated according to predefined criteria, including consistency with current pediatric first aid guidelines, inclusion of essential first-aid management steps, and absence of unsafe recommendations. When minor wording differences occurred, the response that fulfilled these predefined criteria most completely was retained for analysis.

2.6. Expert sample selection and characteristics

The primary data of this study were collected through an expert panel consisting of two independent professionals a nurse and a physician who evaluated the clinical validity and reliability of the information provided by the AI models. Experts declared no conflict of interest with AI model developers.

Experts were recruited using purposive sampling based on their professional experience in pediatric emergency care and first aid management. The experts were not involved in the study design, data collection, statistical analysis, or manuscript preparation. Furthermore, they had no professional, financial, or personal relationship with the developers of the evaluated AI systems. Their role was limited to independently evaluating anonymized AI-generated responses according to predefined assessment criteria.

2.7. Expert qualifications

The experts were selected based on their extensive and up-to-date practical experience in pediatric first aid and emergency management:

- *Expert 1:* A specialist Pediatric Nurse with 20 years of extensive experience in first-line healthcare services and pediatric emergency care, particularly in parental education and counseling. Holds a doctoral degree.
- *Expert 2:* A specialist Physician with 10 years of experience in pediatric emergency and intensive care units, with expertise in managing high-risk clinical conditions and implementing current medical guidelines.

The inclusion of experts with different clinical roles provided a balanced perspective: the nurse contributed insights into practical application and parental clarity, while the physician emphasized clinical accuracy and adherence to emergency protocols.

2.8. Inter-rater agreement

To assess the consistency of the scores provided by the two experts (nurse and physician), Cohen's kappa coefficient (weighted Cohen's kappa for ordinal 5-point Likert data) was calculated. The agreement analysis was performed prior to any consensus discussion or third-expert adjudication, using the original independent ratings of Expert 1 and Expert 2. Based on the evaluation scores for 20 questions across three criteria (accuracy, clarity, comprehensiveness), kappa values were calculated separately for each AI model. Given the ordinal nature of the 5-point Likert scale, a linear weighted kappa approach was applied to account for partial agreement between adjacent response categories. The resulting kappa coefficients were: ChatGPT = 0.43, Claude = 0.51, Gemini = 0.38, and Copilot = 0.47. These values indicate a fair-to-moderate level of inter-rater agreement across models, with Claude demonstrating the highest agreement between experts. In line with established methodological guidelines for subjective clinical evaluations, kappa values between 0.41 and 0.60 are interpreted as moderate agreement, while values below 0.40 indicate fair agreement [14]. The observed level of inter-rater agreement should be interpreted in the context of evaluating open-ended, narrative responses generated by large language models, where variability in clinical interpretation and language expression is expected. Similar AI evaluation studies in healthcare have primarily relied on expert-based Likert scoring without routinely reporting inter-rater reliability coefficients, focusing instead on comparative performance metrics across models [15–17]. In contrast, the present study provides an explicit assessment of inter-rater reliability using Cohen's kappa, thereby strengthening methodological transparency.

In 3 of 20 questions, consensus between the two primary experts was not achieved; therefore, a third independent pediatric healthcare

professional with experience in pediatric emergency care was consulted. The third expert was not involved in any stage of study design, data collection, or analysis. Third-expert adjudication was conducted only after independent ratings were completed and disagreement persisted between the primary evaluators. All assessments were performed under blinded conditions to minimize bias. For statistical analysis, only finalized consensus scores were used. Accordingly, the final dataset for all inferential analyses (including ANOVA and descriptive statistics) was based on a single unified consensus dataset, without duplication or averaging across raters.

2.9. Evaluation protocol

The four AI models (ChatGPT, Claude, Gemini, Copilot) were evaluated by the two experts using a 5-point Likert scale. The assessment was conducted under blinded conditions: responses were anonymized and labeled as "AI Model 1, AI Model 2, AI Model 3, AI Model 4," so the evaluators were unaware of which model had generated each response. This method minimized potential bias in the evaluation.

The scoring focused on three key criteria:

1. **Accuracy (A):** The degree to which the information aligned with medical guidelines.
2. **Clarity (C):** The ease with which parents could understand the information.
3. **Comprehensiveness (Co):** The extent to which critical clinical details and emergency criteria were included (Table 1, Supplementary Section B).

During the evaluation process, both experts independently assessed all criteria. The pediatric nurse primarily focused on clarity and parental clarity, while the physician emphasized clinical accuracy and adherence to emergency care guidelines. However, both experts scored all three criteria to ensure balanced and comprehensive evaluation.

The validity of these criteria was ensured through cross-expert assessment, alignment with medical guidelines, and consistency with clinical practice. The use of blinded evaluation minimized potential bias and enhanced reliability. Independent scoring by the two experts demonstrated repeatability and consistency, supported quantitatively by Cohen's kappa. The evaluation protocol was designed in accordance with international pediatric first aid guidelines and AI evaluation standards (ML model verification standards, AI evaluation metrics). Prior to evaluation, experts underwent a calibration session to standardize scoring criteria and interpretation of the 5-point Likert scale.

2.10. Data analysis

This study involved repeated data collection across three separate days; however, statistical analyses were conducted using consensus-derived scores to minimize pseudo-replication bias and ensure analytical validity. Responses were independently evaluated by a pediatric nurse and a physician using predefined criteria of accuracy, clarity, and comprehensiveness. Inter-rater reliability was assessed prior to consensus procedures using weighted Cohen's kappa for ordinal data. In cases of disagreement, structured discussion was conducted to reach consensus; when disagreement persisted, a third independent pediatric expert provided final adjudication. Only finalized consensus scores were used in all statistical analyses. Descriptive statistics (minimum, maximum, mean, and standard deviation) were calculated for each model and evaluation dimension. Normality was assessed using the Shapiro–Wilk test, and homogeneity of variance was evaluated using Levene's test. As all assumptions for parametric testing were satisfied, one-way analysis of variance (ANOVA) was applied to compare model performance across the four LLMs.Effect size was calculated using partial eta-squared (η^2), and post-hoc pairwise comparisons were performed using Tukey's Honestly Significant Difference (HSD) test to control for multiple testing. A p-value <0.05 was considered statistically significant. The quality of AI-generated responses was assessed based on three dimensions: accuracy, clarity, and comprehensiveness. A standardized evaluation checklist was used for scoring (Table 1, Fig. 1). Because the evaluation instrument used a restricted 5-point ordinal Likert scale and the final dataset was based on consensus-derived scores rather than continuous measurements, score distributions were expected to cluster within a limited number of response categories. Furthermore, the standardized pediatric first-aid scenarios were intentionally developed to represent common, relatively similar levels of clinical complexity. Consequently, limited variability across some item-level scores should not be interpreted as a lack of discrimination between models but rather as a reflection of the structured evaluation framework, the bounded scoring scale, and the use of consensus-based ratings. To enhance transparency, raw expert ratings and the consensus derivation procedure are presented in Table 5.

Table 1
Simple first help questions in children for 5-point likert scale evaluation table.

Score	Accuracy	Clarity	Comprehensiveness
5	Completely correct/ Aligned with guidelines	Very easy to understand for parents	Covers all critical steps and warnings
4	Correct but missing minor details	Understandable with minor jargon	Covers most critical steps
3	Partially correct/ Safe but vague	Moderate difficulty in understanding	Provides only basic information
2	Contains minor scientific errors	Difficult to follow/ Unclear	Inadequate information
1	Completely wrong/ Unsafe	Incomprehensible	Completely inadequate

3. Results

As a result of expert evaluations across three criteria (**Accuracy**, **Comprehensiveness**, and **Clarity)**, the Claude and Gemini models received more consistent and stable scores with lower standard deviations. In contrast, the ChatGPT and Copilot models exhibited wider score ranges and higher deviations among experts, particularly in the **Clarity** and **Comprehensiveness** criteria, indicating greater disagreement. This finding highlights notable differences in response consistency among the models (Table 2). This variability also reflects underlying inconsistency in model performance across repeated evaluations and across different clinical dimensions.

Descriptive statistics were first calculated, including minimum and maximum scores and standard deviations for each model. Claude and Gemini demonstrated low standard deviation values, reflecting higher consistency among expert evaluations. The overall performance of the

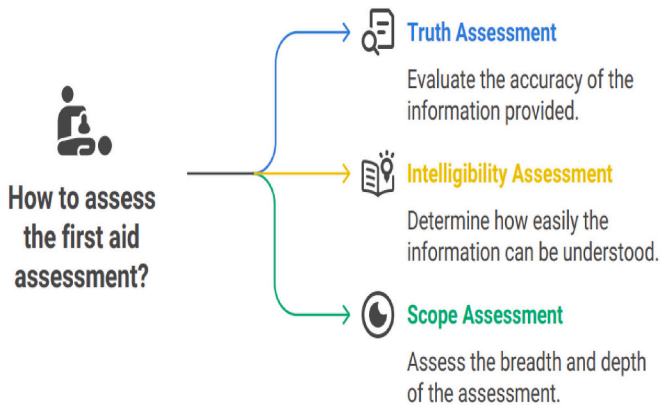


Fig. 1. First aid assessment.

Table 2
Descriptive Statistical Characteristics of AI Model Ratings.

Model	Criterion	Minimum	Maximum	Standard Deviation (SD)
ChatGPT	Accuracy	3.0	5.0	0.65
	Clarity	3.0	5.0	0.71
	Comprehensiveness	2.5	4.5	0.68
Claude	Accuracy	4.5	5.0	0.22
	Clarity	4.0	5.0	0.36
	Comprehensiveness	4.0	5.0	0.30
Gemini	Accuracy	4.0	5.0	0.32
	Clarity	3.5	4.5	0.28
	Comprehensiveness	3.5	4.5	0.26
Copilot	Accuracy	3.0	4.5	0.52
	Clarity	3.0	5.0	0.74
	Comprehensiveness	2.5	3.5	0.58

models was compared using one-way ANOVA, which revealed significant differences ($F(3,76) = 18.72$, $p < 0.001$). To determine the magnitude of these differences, the effect size was calculated using partial eta-squared ($\eta^2 = 0.425$), indicating a large effect size. Claude achieved the highest overall mean score (4.93; 95% CI [4.85–5.00]). Here, 4.93 represents the unweighted grand mean of expert ratings across all items and criteria for the Claude model (5-point Likert scale aggregated across repeated measures). To determine pairwise group differences, a post-hoc Tukey Honestly Significant Difference (HSD) test was performed following the ANOVA to control for multiple comparisons and to identify specific group-level differences. Gemini performed similarly to Claude in accuracy but scored lower in clarity and comprehensiveness, resulting in an overall mean of 4.35. Copilot showed strength in clarity but lower comprehensiveness, with an overall mean of 3.73. ChatGPT ranked lowest, with the lowest averages across all criteria (overall mean = 3.45). Based on expert evaluations across three key criteria (**Accuracy, Clarity,** and **Comprehensiveness),** the average scores obtained reveal significant differences in the response quality of the AI models. The Claude model achieved the highest average scores across all criteria **(Accuracy: 5.0, Clarity: 4.8, Comprehensiveness: 5.0),** with an overall mean score of 4.93, emerging as the most successful model. These results indicate that Claude was rated near-perfect by experts in terms of both information accuracy and content Comprehensiveness. The Gemini model demonstrated strong performance particularly in the **Accuracy** criterion, with an average score of 4.85; however, lower scores in **Comprehensiveness** (4.05) and **Clarity** (4.15) reduced its overall mean to 4.35. This suggests that while Gemini is a robust model in terms of factual correctness, it falls behind Claude regarding clarity of expression and content breadth. The Copilot model, despite achieving a high average score in **Clarity** (4.8), received a low score in **Comprehensiveness** (2.6), resulting in an overall mean of 3.73. This indicates that Copilot's responses were considered understandable by experts but lacked sufficient depth in content. However, this apparent strength in clarity should be interpreted alongside the observed variability in expert ratings and lower comprehensiveness scores, indicating inconsistency in the overall communication quality of Copilot responses.

The ChatGPT model received the lowest average scores across all criteria (Accuracy: 3.75, Clarity: 3.8, Comprehensiveness: 2.8), with an overall mean of 3.45, placing it at the bottom. These results suggest that ChatGPT showed limited success in meeting expert expectations in both information accuracy and Comprehensiveness. Overall, the Claude model emerged as the AI providing the most consistent and high-quality responses, while the performance of the other models varied across criteria. These findings indicate that the choice of AI model in text generation can yield different outcomes depending on the evaluation criteria (Table 3). However, this ranking should be interpreted cautiously, as the study is limited to 20 standardized questions, two expert raters, and a defined data collection period (2024–2025), which may limit the generalizability of the findings to other contexts or time periods. Although Claude consistently outperformed other models in post-hoc comparisons, these results should be interpreted as relative performance differences within the study sample rather than absolute superiority across all contexts.

Table 3
Overall Evaluation Metrics and Scores of AI Models.

Model	Average Accuracy (D)	Average Clarity (A)	Average Comprehensiveness (C)	Overall Average Score
Claude	5.0	4.8	5.0	4.93
Gemini	4.85	4.05	4.15	4.35
Copilot	3.8	4.8	2.6	3.73
ChatGPT	3.75	3.8	2.8	3.45

Note: The descriptive statistics in Tables 2 and 3 are based on repeated measures across three days to ensure response reliability.

3.1. *Superiority of the Claude model: Findings supported by expert ratings*

An examination of the scores from both experts indicates that the Claude model consistently achieved the highest average ratings and demonstrated strong consensus among the evaluators.

In particular, its scores in the accuracy and comprehensiveness criteria reached the maximum observed value of 5.0, suggesting that the model met expert expectations in terms of both informational content and depth of explanation (Fig. 2). Although the Gemini model received scores close to Claude in the accuracy criterion, it was rated lower in comprehensiveness and clarity. This discrepancy may indicate that experts found some of Gemini's responses to be superficial or lacking contextual relevance. Although Copilot achieved relatively higher clarity scores than some competing models, its lower comprehensiveness ratings suggest that understandable language did not always translate into sufficiently detailed clinical guidance. However, its lower scores in the comprehensiveness criterion suggest that the model did not provide sufficient detail or breadth of information for certain questions. The ChatGPT model received the lowest scores from both experts, particularly in comprehensiveness and accuracy, consistently reflecting inadequate performance. These results indicate that its responses to critical health questions were insufficient in both informational accuracy and content breadth. This interpretation is supported by Tukey HSD post-hoc analyses, which demonstrated statistically significant pairwise differences between Claude and all other evaluated models. Overall, the dual-expert ratings highlight Claude as the most successful model in terms of both technical accuracy and communicative adequacy, while the performance of the other models varied across criteria. These findings demonstrate the effectiveness of a multi-expert evaluation approach in distinguishing the strengths and weaknesses of AI models. Nevertheless, the interpretation of Claude's superiority is constrained by the limited sample size, the specific set of pediatric first aid scenarios, and the reliance on a small expert panel, which should be considered when generalizing the results.

Statistical analysis based on expert evaluations revealed significant differences in overall performance among the four AI models (Table 4). Pairwise comparisons were conducted using Tukey HSD post-hoc test, and all reported group differences were adjusted for multiple comparisons. Claude achieved the highest scores within the evaluated sample and demonstrated statistically significant differences compared with the other models in post-hoc analyses. Gemini was significantly higher than Copilot and ChatGPT ($p < 0.01$), but lower than Claude ($p < 0.001$). Copilot and ChatGPT did not differ significantly ($p = 0.24$), indicating statistical equivalence between these two models. Thus, ANOVA alone was not interpreted as evidence of superiority; all conclusions regarding model ranking are based on post-hoc Tukey HSD comparisons (Table 4). Raw expert ratings per model and question (5-point Likert scale). Final consensus scores were used when expert disagreement occurred. Table 5 was added to provide full transparency of the underlying expert evaluations and to allow readers to verify score distributions and inter-rater

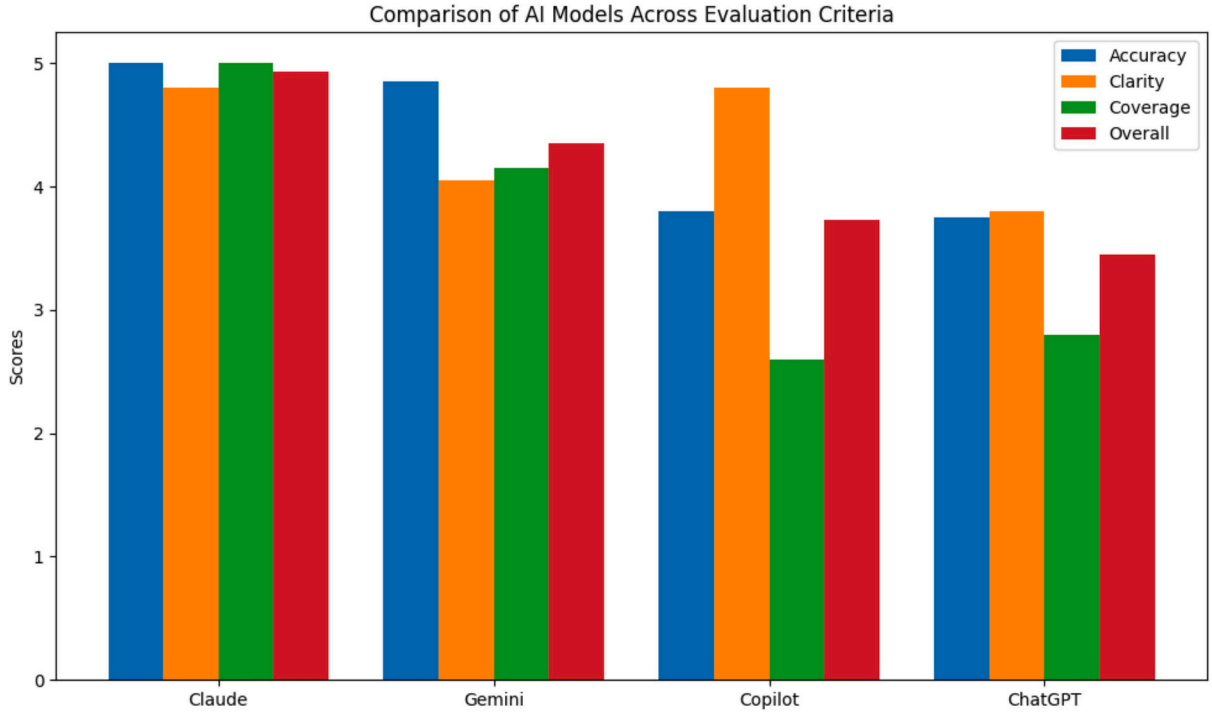


Fig. 2. Comparison of evaluation criteria of artificial intelligence models. Note: Error bars in Fig. 2 represent standard deviations to illustrate variability in expert ratings across three repeated measurements per question over three different days.

Table 4
Overall performance comparison of AI models based on expert evaluations.

Model	Overall Average Score	95% CI	Post-hoc Tukey HSD Comparison	Comment
Claude	4.93	[4.85,5.00]	Significantly higher than Gemini, Copilot and ChatGPT ($p < 0.01$)	Highest overall performance
Gemini	4.35	[4.20,4.50]	Lower than Claude ($p < 0.001$); higher than Copilot and ChatGPT ($p < 0.01$)	Strong factual accuracy
Copilot	3.73	[3.60,3.85]	Lower than Claude and Gemini; not different from ChatGPT ($p = 0.24$)	High clarity but limited scope
ChatGPT	3.45	[3.30,3.60]	Lower than Claude and Gemini; not different from Copilot ($p = 0.24$)	Lowest overall performance

Note: $F(3,76) = 18.72$, $p < 0.001$ indicates statistically significant differences in overall performance. Partial eta-squared (η^2) 0.425 indicates a large effect size. The ANOVA p-value represents only overall group differences and is not interpreted as evidence of pairwise superiority between individual models. Pairwise differences are derived exclusively from Tukey HSD post-hoc comparisons. Pairwise superiority claims reported in this table are based exclusively on Tukey HSD post-hoc comparisons performed after the significant omnibus ANOVA result and not on the ANOVA F-test itself.

variability across questions and models.

4. Discussion

This study systematically evaluated the responses of four large language models (ChatGPT, Claude, Gemini, Copilot) to common pediatric first aid scenarios based on accuracy, clarity, and Comprehensiveness criteria. Our findings align with the existing literature, indicating that while AI models are increasingly accurate, they display variability in response consistency and predictability. The findings of this study indicate that artificial intelligence–based large language models have the potential to provide information in the context of pediatric first aid; however, meaningful differences were observed among the models in terms of clinical accuracy, clarity, and comprehensiveness. Notably, some models achieved high accuracy scores while demonstrating limitations in terms of parental clarity and practical applicability. This highlights that clinically accurate information alone is insufficient; in emergency situations, information must also be conveyed in a manner that allows parents to quickly and correctly understand and apply it. In this context, for time-critical situations such as pediatric first aid, AI-supported systems should be developed with a focus not only on medical accuracy but also on user-centered language, clear guidance, and well-defined safety boundaries.

Notably, ChatGPT occasionally produced comprehensive and creative responses but exhibited high standard deviations, reflecting variability consistent with previous studies. Similarly, Copilot generated explanatory but Comprehensiveness-limited responses, supporting limitations reported in the literature. In contrast, Claude and Gemini demonstrated more stable scores with lower variance, a finding that has been only sparsely reported in prior research. For instance, Claude has shown high coherence in medical decision-making scenarios [15] and strong performance in providing humanistic care during HIV patient consultations [16], supporting its consistent performance. Likewise, Gemini achieved high accuracy in a study but lagged in Comprehensiveness. ChatGPT-4, particularly when enhanced with RAG technology, has shown superior accuracy and clarity [17]. Systematic reviews on healthcare quality management also report that different LLMs offer similar accuracy, with limited significant differences [18]. Together, these findings underscore that not only accuracy but also consistency and comprehensiveness are critical criteria in model selection, reinforcing the importance of multidimensional evaluation frameworks such as the one employed in this study, highlighting the need for multidimensional evaluation in healthcare AI applications. The observed differences between models may be attributable to variations in training

Table 5
Item-level expert ratings and consensus scores across AI models.

Question	Chat GPT E1	E2	Consensus	Claude E1	E2	Consensus	Gemini E1	E2	Consensus	Copilot E1	E2	Consensus
Q1	3.0	4.0	**3.5**	5.0	5.0	**5.0**	4.0	4.0	**4.0**	3.0	3.0	**3.0**
Q2	3.0	3.5	**3.5**	5.0	5.0	**5.0**	4.0	4.5	**4.5**	3.0	3.0	**3.0**
Q3	3.5	4.0	**4.0**	5.0	5.0	**5.0**	4.0	4.0	**4.0**	3.0	3.5	**3.5**
Q4	3.0	3.0	**3.0**	5.0	5.0	**5.0**	4.0	4.0	**4.0**	3.0	3.0	**3.0**
Q5	3.5	3.5	**3.5**	5.0	5.0	**5.0**	4.5	4.5	**4.5**	3.0	3.0	**3.0**
Q6	3.0	3.5	**3.5**	5.0	5.0	**5.0**	4.0	4.0	**4.0**	3.0	3.0	**3.0**
Q7	3.5	4.0	**4.0**	5.0	5.0	**5.0**	4.0	4.5	**4.5**	3.0	3.0	**3.0**
Q8	3.0	3.0	**3.0**	5.0	5.0	**5.0**	4.0	4.0	**4.0**	3.0	3.5	**3.5**
Q9	3.5	3.5	**3.5**	5.0	5.0	**5.0**	4.5	4.5	**4.5**	3.0	3.0	**3.0**
Q10	3.0	3.5	**3.5**	5.0	5.0	**5.0**	4.0	4.0	**4.0**	3.0	3.0	**3.0**
Q11	3.5	4.0	**4.0**	5.0	5.0	**5.0**	4.0	4.5	**4.5**	3.0	3.0	**3.0**
Q12	3.0	3.0	**3.0**	5.0	5.0	**5.0**	4.0	4.0	**4.0**	3.0	3.0	**3.0**
Q13	3.5	3.5	**3.5**	5.0	5.0	**5.0**	4.5	4.5	**4.5**	3.0	3.0	**3.0**
Q14	3.0	3.5	**3.5**	5.0	5.0	**5.0**	4.0	4.0	**4.0**	3.0	3.0	**3.0**
Q15	3.5	4.0	**4.0**	5.0	5.0	**5.0**	4.0	4.5	**4.5**	3.0	3.5	**3.5**
Q16	3.0	3.0	**3.0**	5.0	5.0	**5.0**	4.0	4.0	**4.0**	3.0	3.0	**3.0**
Q17	3.5	3.5	**3.5**	5.0	5.0	**5.0**	4.5	4.5	**4.5**	3.0	3.0	**3.0**
Q18	3.0	3.5	**3.5**	5.0	5.0	**5.0**	4.0	4.0	**4.0**	3.0	3.0	**3.0**
Q19	3.5	4.0	**4.0**	5.0	5.0	**5.0**	4.0	4.5	**4.5**	3.0	3.0	**3.0**
Q20	3.0	3.5	**3.5**	5.0	5.0	**5.0**	4.5	4.5	**4.5**	3.0	3.0	**3.0**

Note: E1 = Expert 1; E2 = Expert 2. Two independent experts evaluated each AI-generated response using a 5-point Likert scale. Consensus scores were derived according to the predefined consensus procedure described in the Methods section and were used in all statistical analyses. The table is presented to provide transparency regarding the underlying expert ratings.

data sources, reinforcement learning strategies, safety alignment mechanisms, and response-generation architectures. Models that prioritize safety and contextual completeness may provide more comprehensive answers but at the expense of conciseness, whereas models optimized for brevity and readability may omit clinically relevant details. These differences highlight the importance of understanding the underlying design characteristics of LLMs when considering their use in healthcare-related information seeking.

Inter-model performance differences have significant clinical implications. Claude's higher scores in accuracy and comprehensiveness suggest that it may provide relatively more reliable responses within the pediatric first aid scenarios evaluated in this study. While Gemini is strong in accuracy, its weaknesses in Comprehensiveness and clarity may result in parents receiving incomplete or superficial critical information. Although Copilot's explanations are understandable, its limited content depth may pose clinical risks due to brief or insufficient recommendations. ChatGPT's lowest performance, particularly in Comprehensiveness, increases the risk of omitting age-specific critical details, potentially elevating morbidity. These findings highlight the need for integrating AI tools with professional clinical oversight and parental guidance interventions to mitigate potential risks. Recent studies support these findings. Kapoor et al. (2025) reported significant differences in accuracy, clarity, and completeness in pediatric dentistry advice provided by AI chatbots [19]. Similarly, ChatGPT-3.5 achieved the highest overall accuracy in general medical diagnosis [20], while Gemini and Copilot performed less effectively. In pediatric orthopedics, ChatGPT and Gemini produced more accurate and comprehensive responses than Copilot [21]. However, Pirkle et al. (2024) noted that these models exhibited limited compliance with clinical guidelines, and Gemini, in particular, occasionally cited inaccurate or fabricated sources [8]. Taken together, these results emphasize that while LLMs hold promise as educational or decision-support tools, they should not replace real-time clinical judgment.

In conclusion, large language models have potential as supportive tools in pediatric first aid; however, they currently contain critical informational gaps. Therefore, improvements in response quality, enhancement of users' health literacy, development of ethical and safety standards, implementation of continuous model performance monitoring, and clinical oversight of these systems are essential.

From a clinical perspective, healthcare professionals should encourage parents to use AI-generated information as a supplementary educational resource rather than a substitute for professional medical advice. Clinicians may also play an important role in improving digital health literacy by helping caregivers evaluate the credibility of online information sources and recognize situations that require immediate medical assessment. Future clinical guidelines may benefit from incorporating recommendations regarding the safe and appropriate use of AI-assisted health information tools.

5. Strengths of the study

This study has several notable strengths that enhance its methodological rigor and contribution to the literature. First, it adopts a novel comparative framework by evaluating multiple large language models (ChatGPT, Claude, Gemini, and Copilot) within a standardized pediatric first aid context, enabling a direct and systematic comparison across leading AI systems. Second, the study employs a multidimensional evaluation approach incorporating accuracy, clarity, and comprehensiveness, which provides a more holistic assessment of model performance compared to studies focusing on a single metric.

Another key strength is the use of a blinded expert evaluation design involving both a pediatric nurse and a physician, which reduces individual bias and increases the validity of the assessment. The inclusion of Cohen's kappa statistics further strengthens the methodological transparency by explicitly reporting inter-rater reliability.

In addition, data were collected across multiple time points, reducing the potential impact of model variability and enhancing response stability. The use of standardized pediatric first aid scenarios grounded in international guidelines (AAP, WHO, and Red Cross) increases the clinical relevance and reproducibility of the study. Finally, the inclusion of item-level data (Table 5) enhances transparency and allows full auditability of the scoring process, supporting reproducibility and methodological robustness.

6. Limitations

This study has several limitations that should be acknowledged. First, the evaluation was based on a limited number of pediatric first aid scenarios and may not capture the full spectrum of emergency situations encountered in real-life clinical and home settings. Second, although expert evaluation was conducted under blinded conditions and supported by acceptable inter-rater reliability, the assessment of open-

ended AI-generated responses inevitably involves a degree of subjective judgment. Third, the expert panel consisted of a pediatric nurse and a physician; however, both experts were affiliated with clinical practice environments where institutional protocols and local practice standards may influence evaluation perspectives. Differences in institutional experience and clinical routines may therefore have had a minor effect on scoring consistency. Fourth, the study did not include parental participants in the evaluation process. Although the primary aim was to assess clinical accuracy and guideline adherence, the absence of parents limits insights into real-world usability, comprehension, and practical applicability of the responses. Fifth, the selection of responses was based on the "most comprehensive and guideline-consistent" output when multiple variations occurred. While this approach was necessary for standardization, it may introduce a degree of selection bias by prioritizing more detailed responses over potentially simpler but practically useful alternatives. Sixth, the use of standardized prompts may not fully reflect the variability, ambiguity, and contextual complexity of real-world parental queries. This artificial structure, while improving methodological control, may limit ecological validity. Seventh, all prompts were submitted exclusively in English, which may limit the generalizability of findings to non-English speaking populations and different cultural or health literacy contexts. Eighth, the study does not evaluate real-world parental behaviors, decision-making processes, or clinical safety outcomes following exposure to AI-generated information. Therefore, findings should be interpreted as performance-based assessments rather than indicators of actual behavioral or clinical impact. In addition, the use of a 5-point Likert scale and consensus-based score aggregation may have reduced score dispersion across individual items, potentially limiting the sensitivity of the evaluation to detect subtle differences between responses. Finally, the models were evaluated using their publicly available versions at a specific time point; therefore, ongoing updates and improvements in large language models may alter performance characteristics over time, limiting temporal generalizability.

7. Conclusion

This study comparatively assessed the performance of large language models in pediatric first aid based on accuracy, clarity, and Comprehensiveness. The findings indicate that models differ not only in accuracy but also in consistency and comprehensiveness. The Claude model demonstrated the highest overall performance, whereas ChatGPT and Copilot showed limitations particularly in Comprehensiveness and stability. These results confirm the potential of AI models as supportive tools in healthcare but highlight that they should not be regarded as standalone reliable information sources in clinical settings. Consequently, the development of AI-based systems should prioritize reliability, consistency, and improvements with clinical oversight.

CRediT authorship contribution statement

Birsel Molu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethical approval

This study did not require ethical committee approval, as it exclusively utilized responses generated by ChatGPT, Claude, Gemini, and Copilot, and did not involve direct human participation. The AI models employed are advanced language models capable of providing responses to academic and clinical inquiries. The responses obtained in this study were evaluated as tools for academic research, and their accuracy was corroborated using reliable scientific sources and up-to-date pediatric first aid guidelines.

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Declaration of competing interest

This study has not been published anywhere before.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ienj.2026.101885.

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